INTERNATIONAL INSTITUTE OF PROFESSIONAL STUDIES,

DEVI AHILYA VISHWAVIDYALAYA, INDORE

MCA 5 YEAR (BCA 3Yrs + MCA 2Yrs)

PROGRAMME

1 Semester

Scheme for MCA 5 year (BCA 3 Yr + MCA 2 Yr)

INTERNATIONAL INSTITUTE OF PROFESSIONAL STUDIES

DEVI AHILYA UNIVERSITY, INDORE

MCA (5 Years)

I SEMESTER

July- 2023 – December – 2023

Code	Subject	L	T	P	C
IC-101A	Mathematics I	3	1	0	4
IC-104C	Problem solving using 'C'	3	1	0	4
IC-105	PC Software	3	1	0	4
IC-103B	Digital Electronics	3	1	0	4
IC-106C	English and Communication Skills	3	1	0	4
IC-107E	Problem solving using 'C' Lab	0	0	4	2
IC-111B	Digital Electronics Lab	0	0	4	2
IC-110C	PC Software Lab	0	0	4	2
IC-108	Comprehensive Viva	0	0	0	4
Total					30

MCA 5 Years -2024
Semester - I
Subject Name: Mathematics-I

Course Type: - Core

Course Credits – 4-Theory: 0-Practical

Course Objective –To provide a course on elementary mathematical techniques and familiarize students with basics of differentiation and integral calculus.

Course Outcomes – After completion of this course, learners will be able to

- **CO1-** Understand basic concepts of Partial differentiation, Maxima & Minima of the function, convergence and divergence of the series.
- **CO2**–Solve mathematical problems based on the course material.
- **CO3–**To develop mathematical skills and methods appropriate for students in the computer science.
- **CO4-**To prepare students for more advanced mathematical courses.

Course Content–

Unit No	Title	Contents	Hr	Targeted Levels of Blooms T. (Q1)	Content and Pedagogy (Q2)	(Q3) Online Resources	(Q4) Assign./ Assessment/Discussion
1	Differential Calculus - I	Successive differentiation, Leibnitz's theorem, Expansion of functions, Maclaurin's theorem, Taylor's theorem, **LO1:**Apply Leibnitz's Theorem, Maclaurin's theorem and Taylor's theorem,	7	1, 2, 3	Book 2, 3 (Lecture)	https://testbook.com/maths/differential-calculus	Assign: questions on Leibnitz's theorem and Expansion of functions.

2	Differential Calculus - II	Indeterminate forms. Tangents and Normal, Curvature, Asymptotes. **LO1:** Understanding Concepts of Tangents and Normal, Curvature, Asymptotes.	7	1, 2, 4	Book 2, 3 (Lecture)	https://mathworld.wolfram.com/Curvature.html	Discussion on Indeterminate Forms and Curvature.
3	Partial Differentiation	Euler's theorem on homogeneous functions, Mean value theorem and Taylor's theorems of two variables. **LO1:** Apply theMean value theorem and Taylor's theorems.	6	1, 2, 3	Book 2, 4 (Lecture)	https://testbook.com/maths/homogeneous-function	Questions on Euler's and Mean Value Theorems.
4	Application of Partial Differentiation	Maxima and minima of functions of two and more variables, Lagrange's method of undetermined multipliers. **LO1:** Using the partial derivatives in optimization, social sciences, physics, Life Sciences etc.	6	1, 2, 3, 4	Book 3, 4 (Lecture)	https://www.iist.ac.in/sites/default/files/people/Lagrange-Multiplier.pdf	Assign: Practice and Assignment questions on Maxima and Minima.
5	Integral Calculus - I	Integration of irrational and Transcendental functions, Reduction formulae **LO1:**Apply the Reduction Formulae for Various Integrals.	7	1, 3, 4	Book 1 (Lecture)	https://www.cuemath.com/reduction-formula/	AssignmentQuestions on Integration and Reduction Formulae.
6	Integral Calculus - II	Integral as the limit of a sum, summation of series. **LO1:**To find Summation of Series using Integrals.	5	1, 3, 5	Book 1 (Lecture)	https://www.toppr.com/guides/maths/integrals/definite-integral-as-a-limit-of-a-sum/	Questions on Summation of Series.
7	Convergence and Divergence	Convergence and Divergence of infinite series, Definition and various tests. **LO1:** Apply the various tests to determine Convergence and Absolute Convergence of an	7	1, 2, 3, 4	Book 4 (Lecture)	https://germanna.edu/sites/default/files/2022-09/Converging%20and%20Div	Discussion on various tests for Convergence and Divergence of infinite series.

		Infinite series of Real Numbers.				erging%20Serie s.pdf	

Books and Reading:

1. Gorakh Prasad, Integral Calculus.
2. Gorakh Prasad, Differential Calculus
3. Shanti Narayan, Differential Calculus.
4. Devi Prasad, Advanced Calculus.

CO-PO Mapping

	PROGRAM SPECIFIC OUTCOMES	CO1	CO2	CO3	CO4
PSO1	Algorithm development	✓			
	Computer Based Solution			✓	
	Application Development				✓
	Disciplinary Knowledge	✓	✓	✓	
PSO2	Programming Skill				✓
	Problem Solving/ Critical thinking	✓		✓	

	Mathematical Analysis	✓	✓	✓	
PSO3	Financial Aspects	✓			
	Modern Tools			✓	
	Complex Problems solving	✓		✓	
	Time Frame Analysis				
PSO4	Computer Architecture				
	Cooperation/Teamwork				
	Resource Management				
PSO5	Modern Problem solving Technique	✓	✓	✓	
	Business Skills				
PSO6	Communication skills				
	Decision making skills	✓			✓
	Managerial Skill				
	Environmental constraints				

	Research Skills				

MCA (5Yrs)-I SemesterIC- 103B

Digital Electronics

Prerequisite:-Knowledge of Basic Electronics.

Course Type: - Core

Course Credit:- 4 (Theory)

Course Objective:-To present a problem oriented introductory knowledge of Digital circuit and its applications and tofocus on the study of electronic circuit.

Course Outcomes:-

CO1:-Examine the structure of number systems and perform the conversion among different number system.

CO2:- Realize combinational circuits for given application

CO3:-Design and analyses synchronous and asynchronous sequential circuit using flip-flop.

CO4:-Illustrate reduction of logical expressions using Boolean algebra, K-map and tabulation method and implement thefunction using logic gates.

Course Content:-

Unit No.	Name	Contents	Hours	Targeted Levels	Content and Pedagogy	Online Resources	Assign/Assessment/Discussion
1	Number system and codes	Binary,octal,hexadecimal,decimal number system and their inter conversion,BCD number(8421,2421),gray code,excess-3 code.binary addition and subtraction,signed and unsigned binary number,1's and 2'scomplement representation	5	1 and 2			
	LO1:-Upon completion of this unit students will be able to identify and apply number systems operations and codes						
2	Boolean Algebra and logic Gates	Basic logic circuit,logic gates(AND,OR,NOT,NAND,NOR,EX-OR,EX-NOR and their truth tables) Laws of Boolean algebra,De-Morgan's theorem,Minterm,Maxterm,POS,SOP,K-Map,Simplification by Boolean theorems,don't care condition	5	1 and 2			
	LO2:-Define different types of logic gates,identify their Ics and also verify their truth table and iiiustrate realization of boolean expression in SOP and POS form and design it using logic gates						
3	Logic Families	Introduction to digital logic family such as RTL, DTL, TTL etc. Their Basic circuit,performance characteristics	6	1 and 2			
	LO3:-Upon completion of this unit the student will able to identify digital logic family						
4	Arithmetic Circuits	Half adder,Fulladder,Half subtractor,Full subtractor,parallel binary adder,controlled inverter,	6	1,2 and 3			
	LO4:-Derive basic logic gates adder and subtractor and also derive arithmetic circuits using universal gates						
5	Combinational Circuits	Multiplexers,Demultiplexers,Decoders,Encoders,Parity generators/Checkers,Code converters,	6	1,2 and 3			
	LO5:-Design and test combinational circuits						

6	Flip-Flop and Timing circuit	Set-Reset latches,RS flip flop,D-flip flop,JK flip flop,T-flip-flop	6	1,2,3 and 4			
	LO6:- Design and develop sequential circuits Flip-Flop						
7	Registers and Counters	Synchronous/Asynchronous counter, up/down counter, shift registorsr,Bi-directional register	6	1,2,3 and 4			
	LO7:- Design and develop sequential circuits Register and Counter						

***1- Remember, 2-Understand,3-Apply,4-Analyze**

Text Books:-

1 Digital Fundamental by Morris and Mano,PHI publication

2 Fundamental of Digital circuits by A.Anand Kumar ,PHI publication3 Modern Digital Electronics by

R.P. Jain,prentice hall of india

4 Digital Circuits and Design by S.Salivahanan and S.Arivazhagan5 Digital Principal and

Applications,Malvino and Leach,TMH

6 Digital Fundamentals, Thomas L.Floyd,Pearson Education

CO-PO Mapping

	PROGRAM SPECIFIC OUTCOMES	CO1	CO2	CO3	CO4
PSO1	Algorithm development			√	√
	Computer Based Solution	√		√	
	Development Application	√	√	√	√
	Disciplinary Knowledge	√	√		
	Environmental constraints	√		√	
PSO2	Programming Skill				
	Problem Solving/ Critical thinking	√	√	√	√
	Mathematical Analysis	√		√	√
PSO3	Financial Constraint				
	Modern Tool	√	√	√	√
	Complex Problems	√	√		√
	Meet Time Frame	√		√	
PSO4	Computer Architecture	√	√	√	√
	Cooperation/Teamwork		√	√	
	Resource Management			√	√

PSO5	Modern Problem solving Technique	√	√	√	√
	Business Skills	√	√		
PSO6	Communication skills				
	Decision making skills	√			√
	Managerial Skill	√		√	
	Environmental constraints		√	√	
	Research Skills	√	√	√	√

International Institute of Professional Studies, DAVV
MCA (5Yrs) , Semester Ist
Subject Name: Problem Solving Using C (IC-104B)

Course Type :- Core

Course Credits –Theory -4

Course Objective – To develop logic of problem solving and learn basics of methodologies for C language programming.

Course Outcomes – After completion of this course, learners will be able to

- CO1- Develop the logic for the given problem which will help to develop programs, applications in C and gain experience of procedural language programming.
- CO2– Recognize and understand the syntax and construction of C code and Understand functional hierarchical code organization along with arrays, character and strings.
- CO3- Understanding a defensive programming concept and ability to handle possible errors during program execution.
- CO4 - Easily switch over to any other language in future.

Course Content –

Unit No	Name	Contents	Hours	Targeted Levels (Q1)	Content and Pedagogy (Q2)	Online Resources (Q3)	Assignment/ Discussion (Q4)
1	Introduction to Programming Language & Problem solving Approach	Development of flow charts & Algorithms, Why Programming Language? Program development steps, Programming language classification, Translators, Program design techniques.	5	Remember, Understand	(Lecture, discussion)		Draw flowcharts & algorithms for various given problems.
LO: Acquisition of clarity about concepts of Programming languages.							
2	Introduction to C language	History of C Language, Features of C Language, Why is C Language Popular? Structure of C Program, A Sample C Language Program. Errors, Compilation and Execution of C Programs and Exercises.	5	Remember, Understand	(Lecture, discussion)		Write programming codes to understand the structure of C programs.

LO: Acquisition of clarity about fundamentals of C Programming.							
3	Useful terms of Language/Basic Concepts of C	Data types, The C character set, Constants, Variables, Keywords, C Instructions, Type Modifier, Operator Expressions and Assignment Statements : Arithmetic Operators, Relational and Logical Operators, Increment and decrement Operators, Assignment Operators and Expressions, Conditional Expression, Precedence and order of Evaluation and Exercises.	6	Understand, Analyze	(Lecture, Exercises & discussion)		Write C programs to make use of various Datatypes, Operators & Constants.
LO: Illustrate the key concepts of C language required for programming.							
4	Input/output functions and Control Structures	I/O functions, Decision Control Structures, Loop Control Structures, Case control structures, break and continue Statement, Conditional Statements and Exercises.	6	Analyze, Evaluate, Create	(Lecture, Exercises & discussion)		Write C programs to make use of I/O functions, Decision Control & Loop Control Structures and Conditional Statements.
LO: Understood the various control & loop structures with the help of programming exercises.							
5	Arrays and String Manipulation	Introduction to Arrays, One Dimensional Array, Multidimensional Array, Initialization, Declaration, Storage and Access Mechanisms on Array and Exercises. Introduction to Strings, Standard library string functions, Two Dimensional Array of characters.	6	Apply, Evaluate, Create	(Lecture, Exercises & discussion)		Write C programs to make use of Arrays & Strings.
LO: Understood the concepts of Arrays, Strings and various string functions with the help of programming exercises.							
6	Functions	Introduction to Functions, Function Declaration and Prototypes, Function Definition, Call by Value and Call by Reference, return statement, exit() function, Function with arguments, Calling Function with Array, Recursion in Function, Introduction to pointers.	6	Apply, Evaluate, Create	(Lecture, Exercises & discussion)		Write C programs to make use of User defined Functions & Pointers.
LO: Understood the various concepts of functions with the help of programming exercises.							

7	Structure & Union	Structure Definition, Giving Values to members, Structure initialization, Comparison of Structure variables, Array of Structure, Array within Structures, Structures within Structures, Passing Structures to Functions, Why use Structure, Features and Uses of Structures Union Definition and Declaration, Accessing a union Member, Union of Structures, Initialization of a Union Variable, Use of Union, Use of User Defined Type Declarations	6	Apply, Analyze, Evaluate, Create	(Lecture, Exercises, Presentations & discussion)		Write C programs to make use of Structures & Unions.
LO: Understood the various concepts of Structures & Unions with the help of programming exercises and get skilled for developing programs for complex problems.							

Text Book:

1. Y.P. Kanitkar, Let us C, B.P.B. Publication

Reference Books:

1. C -The Complete Reference, Tata Mcgraw Hill
2. Deitel & Deitel, C-How to Program.

Online Resources (Q3):

1. https://nptel.ac.in/cources.
2. https://swayam.gov.in/explorer

CO-PO Mapping

	PROGRAM SPECIFIC OUTCOMES	CO1	CO2	CO3	CO4
PSO1	Algorithm development	√	√	√	√
	Computer Based Solution	√	√	√	√
	Application Development	√	√	√	√
	Disciplinary Knowledge	√	√	√	√
PSO2	Programming Skill	√	√	√	√
	Problem Solving/ Critical thinking	√	√	√	√
	Mathematical Analysis	√	√		√
PSO3	Financial Aspects				
	Modern Tools			√	√
	Complex Problems solving	√	√	√	√
	Time Frame Analysis	√			√
PSO4	Computer Architecture				
	Cooperation/Teamwork	√	√	√	√
	Resource Management	√	√	√	√
PSO5	Modern Problem solving Technique	√	√	√	√
	Business Skills				
PSO6	Communication skills				
	Decision making skills	√	√	√	√
	Managerial Skill				√
	Environmental constraints				
	Research Skills				√

MCA 5Yrs

Semester I

Subject Name: English & Communication Skills (IC-105)

Course Type :- Core

Course Credits – 4-Theory

Course Objective –

- To impart knowledge about importance of effective communication in personal as well as professional life.
- To improve their communication and linguistic competence in English language.
- To enable students to become proficient communicators by explaining the dynamics of communication.

Course Outcomes – After completion of this course, learners will be able to

CO1: To analyze the forms of communication and make the students identify and interpret the verbal and non-verbal communication in order to choose their body language more correctly and impressively in accordance with organizational requirement.

CO2: To develop their problem solving skills, interpersonal skills, presentation skills and professional skills for handling day-to-day managerial responsibilities.

CO3: To gain confidence and help them to transform their communication skills and decision making ability to communicate in daily situations effectively.

CO4: To build professional conduct and enhance the skills of students in written as well as oral communication and to make them translate and evaluate the strategies of effective business writing.

Course Contents -

Unit No	Title	Contents	Hrs	Targeted Levels of Bloom's T.(Q1)	Content and Pedagogy(Q2)	(Q3) Online Resources	(Q4) Assign./Assessment/Discussion

1	**Introduction to Communication**	Meaning and Definition of communication, Process of Communication, Importance of Feedback, functions of communication. **LO:** Define and explain basic concepts of communication skills.	4	1*	Study Material from the textbooks (Lecture and case study)	#	Discussion through case study.
2	**Objectives of effective Communication**	Information, order, advice, suggestion, motivation, persuasion, warning, education, raising morale. **LO:** Outline, identify and compare various objectives of communication in business organization.	5	2,3,4 and 5*	Study Material from the textbooks (lecture and discussion)	#	Assignment on actual objectives .
3	**Effective communication**	Components of effective communication, 5W's & 1H & 7 C's of effective communication, Communication Barriers, Listening- Importance of Listening, Types of Listening, activities to develop listening skills, Barriers to Listening and overcoming them. **LO:** Develop listening skills among students and help them overcome barriers to effective communication.	6	6*	Study Material from the textbooks (lecture and case study)	#	Discussion through Case study
4	**Media of Communication**	Verbal and Non-verbal: Importance of non-verbal communication, Kinesics, Proxemics, Paralanguage , written and oral communication **LO:** Able to use, categorize and correlate about verbal and non-verbal forms of communication.	7	3 and 4*	Study Material from the textbooks (lecture and discussion)	#	Assignment on Actual medias of communication

5	**Channels of Communication**	Formal and informal channels of communication, Dimensions of communication: Upward, downward, lateral/horizontal, diagonal, Consensus, Informal Communication: Grapevine- Advantages and disadvantages. **LO:** Compare, contrast and Appraise various formal and informal channels of communication in Business organization using Organizational flowchart.	6	2,3,,4 and 5*	Study Material from the textbooks (Organizational flowchart)	#	Assignment on Organizational flowchart
6	**Business Correspondence**	Report writing, letter writing : Basic patterns of business letters. **LO:** Able to plan, write and develop professional writing skills using different letters and reports of firms.	5	5 and 6*	Study Material from the textbooks (Letter and report formats)	#	Assignment on Written Business correspondence
7	**Presentation Skills**	Making effective presentation, Speeches and Public speaking, Group discussion, Preparing for Interviews **LO:** Interpret, apply and develop presentation skills using group discussion and presentation.	7	2,3 and 6*	Study Material from the textbooks (GD,Presentation)	#	Assignment on Presentation making and GD practice.

***1-Remember, 2-Understand, 3-Apply,4-Analyze,5-Evaluate,6-Create**

 # Online Resources will be shared during classes.

Books and Reading:

1. C. S. Rayudu, Communication.
2. Andal N., Communication Theories and Models.
3. Keval J. Kumar, Communication Barriers.
4. Denis McQuail, Effective Communication.

5. K.K.Sinha, Business Communication
6. Asha Kaul, Effective Communication
7. M. V. Rodriques, Effective Business Communication.

CO-PO Mapping

	PROGRAM SPECIFIC OUTCOMES	CO1	CO2	CO3	CO4	CO5
PO1	Algorithm development					
	Computer Based Solution					
	Application Development					
	Disciplinary Knowledge	✓	✓	✓	✓	✓
PO2	Programming Skill					
	Problem Solving/ Critical thinking	✓	✓	✓	✓	✓
	Mathematical Analysis					
PO3	Financial Aspects					
	Modern Tools					
	Complex Problems solving		✓			
	Time Frame Analysis					
PO4	Computer Architecture					
	Cooperation/Teamwork	✓	✓	✓	✓	✓
	Resource Management					
PO5	Modern Problem solving Technique		✓			
	Business Skills	✓	✓	✓	✓	✓
PO6	Communication skills	✓	✓	✓	✓	✓
	Decision making skills		✓	✓		
	Managerial Skill	✓	✓	✓	✓	✓
	Environmental constraints	✓	✓	✓	✓	✓

	Research Skills	✓	✓	✓	✓	✓

MCA 5 Yrs
Semester - I
Subject Name – PC Software (IC - 105)

Course Type: - Core
Course Credits – 4 Theory 2 Practical

Course Objective – To familiarize students with the core concepts and practices of computers and its working. To know the practical application of various automation tools like Microsoft Office environment, Web browser, Multimedia tools, etc.

Course Outcomes – After completion of the course, student will;

CO1: - To acquire clarity of the basic components and the design of CPU, ALU and Control Unit.

CO2: – To classify memory hierarchy and its concurrent access to the memory system, to train students thoroughly in different number system and their conversion.

CO3: – To correlate software identifies software types like Operating Systems and DBMS. Demonstrate detailed understanding of Microsoft Office environment.

CO4: – To use concepts of Networking, the Internet, and different terms related to the Security, Cryptography and Multimedia.

Course Content -

Unit No	Name	Contents	Hours	Targeted Levels of Blooms T. (Q1)	Content and Pedagogy (Q2)	Online Resources (Q3)	Assignment/Assessment/Discussion (Q4)
1	Introduction to Computer	Definition, Characteristics, Functions and applications of Computer, Components of a Computer: Hardware and Software. Block diagram of a computer: Input devices, Output devices, CPU, Memory. Classification of computer, Generation of computer. **LO:** Comprehend the definition, characteristics, functions, and	6 Hrs	1 and 2*	Ch. 1-4 from Alexis Leon & Ch. 1-2 from D.P. Nagpal	Some useful web link given below**	Assignment/Assessment given below***

		applications of a computer, elaborate its hardware and software components, interpret the input/output devices, CPU, and memory, and classify computer generations.			(Lecture and discussion)
2	Data Representation	Number system - Binary, Octal, Decimal, Hexadecimal and its conversion. Computer software: System software and Application software. Computer languages: Machine, Assembly, High level and Fourth generation languages. **LO:** Demonstrate a comprehensive understanding of number systems and their conversions differentiate between system software and application software, and identify the characteristics and differences of computer languages.	6 Hrs	1, 2, 3 and 6*	Ch. 3, 8 from D.P. Nagpal & Ch. 10, 12 from P.K. Sinha (Lecture, discussion and numerical exercise)
3	Introduction to Operating System & DBMS	Definition and functions of an Operating System, Type and classification of Operating Systems. Introduction to Data Base Management System: Introduction, Quality of information, What is Database, DBMS? Why a database, DBMS? Types of DBMS. **LO:** Define and explain the functions of an Operating System, distinguish between various types and classifications of Operating Systems, understand the fundamentals of DBMS, evaluate the quality of information, and comprehend the significance and types of DBMS in the context of managing databases efficiently.	6 Hrs	1 and 2*	Ch. 10 from D.P. Nagpal & Ch. 29 from Alexis Leon (Lecture and discussion)

4	Microsoft Office Environment - I	Microsoft Word: Working with Word, Typing and Editing, Formatting Text, Page design and layout, Adding tables, Using graphs, Mail merge. Microsoft Excel: Working with excel, Entering data, formatting, Customizing workplace, Calculation in worksheet, Adding charts and Explore features of excel. **LO:** Proficiently navigate and utilize Microsoft Word for typing, editing, text formatting, page design, table insertion, graph creation, and mail merge, as well as demonstrate proficiency in Microsoft Excel by effectively entering data, formatting cells, customizing the workplace, performing calculations in worksheets, adding charts, and exploring various features of Excel.	6 Hrs	1, 2, 3 and 6*	Ch. 6-11 from Sanjay Saxena (Lecture, discussion and creation)
5	Microsoft Office Environment - II	Microsoft–PowerPoint: Working with PowerPoint, Adding Text, Including Multimedia, Customize PowerPoint. Microsoft Access: Creating database, addition and deletion of records, searching, sorting and indexing the records, Creating tables and records, Explore features of Access. **LO:** Skillfully uses Microsoft PowerPoint to create presentations, add text and multimedia elements, and customize PowerPoint features, as well as demonstrate proficiency in Microsoft Access by creating databases, managing records through addition, deletion, searching, sorting, and indexing, creating tables, and effectively exploring various features of access.	6 Hrs	1, 2, 3 and 6*	Ch. 12-13 from Sanjay Saxena (Lecture, discussion and creation)
6	Internet and World Wide Web	Introduction, Internet access, Internet basics, Internet protocols, Internet addressing, Web pages and HTML, Web browser and search engines, Electronic mail. Computer Security: Physical access restriction, Passwords, Firewalls, Cryptography, Computer virus, Bombs and worms. Antivirus software. Introduction	6 Hrs	1 and 2*	Ch. 18 from P.K. Sinha & Ch. 23-26 from Alexis

		of MSDOS. **LO:** Demonstrate a comprehensive understanding of the Internet, including its web browsers, search engines, and electronic mail; possess knowledge of computer security measures such as physical access restriction, passwords, firewalls, cryptography, and computer viruses as well as the use of antivirus software.			Leon (Lecture and discussion)
7	Introduction to Multimedia	Introduction, Multimedia in entertainment, Multimedia in software training, education training etc., Multimedia server and databases, Multimedia tools. **LO:** Explain the fundamental concepts of multimedia, its applications in entertainment and software training, as well as its implementation in multimedia servers and databases, and gain proficiency in using various multimedia tools for creating and manipulating multimedia content.	4 Hrs	1 and 2*	Ch. 19 from P.K. Sinha & Ch. 36-37 from Alexis Leon (Lecture and discussion)

*** 1-Remember, 2-Understand, 3-Apply, 4-Analyze, 5-Evaluate, 6-Create**

Books and Reading:

1. Alexis Leon, Introduction to Computer, Leon Press and Vikas Publications.
2. Alexis Leon, Introduction to Information Technology, Leon Press and Vikas Publications.
3. P.K. Sinha, Fundamentals of Computers, BPB Publications.
4. D.P. Nagpal, Computer Fundamentals, S. Chand Publications.
5. Sanjay Saxena, M.S. Office for Everyone, Vikas Publications.
6. Peter Norton's, Introductions to Computer, Galgotia Publications.
7. R.K. Taxali PC Software for Windows Made Simple, Tata McGraw Hills, New Delhi.

****Online Resources (Q3):**

1. https://nptel.ac.in/courses

2. https://swayam.gov.in/explorer
3. https://ncert.nic.in
4. https://www.geeksforgeeks.org
5. https://www.nielit.gov.in
6. https://swayamprabha.gov.in/
7. https://www.education.gov.in/ict-initiatives

*****Assignment/Assessment/Discussion (Q4):**

Assignment - 1 MS-Office -Word

Assignment -1.1

1. Format text color, bold, and size at least 75% of the time
2. Insert a file INTO an existing Word document
3. Format text into columns
4. Insert a picture from Clip Art and the Design Gallery Live at least 75% of the time
5. Change text wrapping around a picture at least 75% of the time
6. Apply borders and shading to a whole page using the Format Borders and Shading command

Assignment -1.2

Open a blank Microsoft Word document. You can use Microsoft ClipArt, or Clips Online, to do the followingpractice exercises.

Insert a picture of a sun or sunset - Use Format->Size to resize the picture to 1.5" wideUse In-Line Text Wrapping.

Next to the picture type - The weather is great!

Insert a picture of a camera - Change the Text Wrapping to Tight Resize the picture to be 2.5 inches tall Place the picture to the bottom of the page.

Insert a picture of a beach - Format Text Wrapping to Tight Place the picture into the center of the page Add a thick BLUE border around the pictureCrop the picture 5 inches from the left.

Save your practice document and name it - Beginning Word Practice 2<your name>

Assignment -1.3

Create a Letter - Example below:

Type the company name and address

Open a blank Microsoft Word document. Type the following information:

Indian house Academy, 8923 Park dale, New Delhi, BC, V9B 4G9, 474-5311

Select all of the text and use the Font options to format the type:Tahoma, 12 point, bold, centered, and dark red

Select the first line of type and make it 14 point.

Insert a Picture from ClipArt - Search for a photo or cartoon of a light house Select an image and Download it. Use one of the pictures for a company logo Resize the picture Center it above the Company name and address.

Insert the Date and Time - Remember, the default Date and Time updates automatically. This option is not appropriate for medical or legal documents that must be date/time stamped, but is fine for this exercise.

Type a sample business letter:
Dear Mr. Chalifour,

Write in complete sentences and in paragraph form 10 things you like about Lighthouse ChristianAcademy.

Sincerely, Your Name

Assignment -1.4

4.1 Create a Resume.

Assignment - 2

MS-Office - Excel Spreadsheets

Exercise 1.1 – Creating a Spreadsheet
Exercise 1.2 – Changing the look and style
Exercise 1.3 – Adding Formulae
Exercise 1.4 – Adding a Row to a Spreadsheet
Exercise 1.5 – Making a Graph of the Data
Exercise 1.6 – Multiple Sheets

Exercise 1.7 – Dynamic Linking & Explore More Advanced Excel Functions Advanced Word Techniques
Exercise 2.1 – Building a Word Template
Exercise 2.2 – Updating Normal.dot
Exercise 2.3 – Inserting a Table of Contents
Exercise 2.4 – Inserting Cross-References Composite Documents
Exercise 3.1 – Producing a Spreadsheet
Exercise 3.2 – Making a Graph of the Data
Exercise 3.3 – Making a Drawing
Exercise 3.4 – Adding a Picture to the Document
Exercise 3.5 – Adding a Table to the Document
Exercise 3.6 – Adding a Graph to the Document
Exercise 3.7 – Adding an Object

Assignment - 3

MS-Office - Power point presentation

Assignment – 3.1

Add new slides at least 75% of the time, Enter and edit text in a slide at least 75% of the time. Insert a Text Box at least 75% of the time, Format the fill and border of a Text Box, Change text direction and Text alignments in Text boxes, Format text size, font face, color, and bold at least 75% of the time Format Slide background.

Assignment – 3.2

The Learner will be able to: Insert and Format Slide Text. Insert Picture from Clip Art at least 75% of the time, Format picture using Picture Tools at least 75% of the time, Insert an AutoShape, Format AutoShape color and lifestyle at least 75% of the time, Group and move Objects.

Assignment – 3.3

Create a new PowerPoint using a Design template. Insert and Format pictures from ClipArt of from Files at least 75% of the time. Use and modify animations at least 75% of the time, Add Sound to Custom Animation Effects. Insert slide transitions and modify the timing, View the Slide Show at least 75% of the time.

Assignment – 3.4

Change the View to Slide Master at least 75% of the time. Use the Slide Master to change the text formatting at least 75% of the time, add an image to the Slide Master at least 75% of the time. Modify the Slide Background, Edit the Footer. Close the Slide Master and Return to the Normal View at least 75% of the time, Add sample text and review the slide design.

CO-PO Mapping

PSO	Program Specific Outcomes	CO1	CO2	CO3	CO4
PSO1	Algorithm development				
	Computer Based Solution		√	√	
	Application Development				
	Disciplinary Knowledge	√	√	√	√
PSO2	Programming Skill				
	Problem Solving/ Critical thinking		√	√	√
	Mathematical Analysis		√		
PSO3	Financial Aspects			√	√
	Modern Tools		√	√	√
	Complex Problems solving			√	
	Time Frame Analysis				
PSO4	Computer Architecture	√	√	√	√
	Cooperation/Teamwork		√	√	√
	Resource Management	√	√	√	√
PSO5	Modern Problem solving Technique			√	√
	Business Skills			√	√
PSO6	Communication skills	√	√	√	√
	Decision making skills		√	√	√
	Managerial Skill				
	Environmental constraints				
	Research Skills				